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Social media engagement and real-time marketing: Using net-effects and set-theoretic approaches to understand audience and content-related effects

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Abstract

This study examines how various audience-related variables and distinct real-time marketing contents (i.e., predictable and unpredictable real-time marketing) affect social media engagement. The relations are tested using Structural Equation Modeling (SEM) and fuzzy-set qualitative comparative analysis (fsQCA), with a convenience sample that was obtained via conducting three online consumer surveys for several different brands. The SEM findings show that intensity of social media usage, self-brand congruence, and brand-real-time marketing moment congruence are all significant in explaining total engagement. Only self-brand and brand-moment congruence are relevant with predictable real-time marketing content, whereas only brand-moment congruence is significant with unpredictable real-time marketing. On the other hand, the FsQCA findings show various conditional configurations for both the presence and absence of engagement in each content context, where the individual variables need to be combined with others. The results also show that brand-moment congruence is more important to explain engagement with unpredictable content in comparison to other variables, and that differences thus exist with distinct content strategies. This research enriches engagement and real-time marketing lit.erature and the findings can assist content managers in the selection of social media content and in building more profitable consumer-brand relations.

KEYWORDS

brand-moment congruence, FsQCA, involvement, real-time marketing, self-brand congruence, social media engagement, structural equation modeling

1 | INTRODUCTION

Consumers increasingly connect with each other through Web 2.0 technologies and social media platforms. According to Statista (2021), both Northern and Western Europe registered a social media penetration rate of 79% in 2021, followed closely by Northern America with 74%. Globally, the average penetration rate was 53.6% in 2021. Social media platforms have now become part of consumers' daily lives and businesses are eager to take advantage of social media marketing at a strategic level (Felix et al., 2017). These platforms not only enable the consumption and sharing of content among consumers, but also provide information about brands, their products, and their services (Voorveld, 2019).

The increasing use of social media by consumers to keep up to date with news, trends, and events has led brands to look for new ways to feature more prominently on these platforms, namely by communicating at the right moment with real-time marketing (Willemsen et al., 2018). Real-time marketing (or RTM) consists of incorporating news, events, and trends happening in the moment into the social media content of brands to capture attention and drive positive consumer responses (Mazerant et al., 2021). This contrasts with normal brand content, which does not depend on topical and timely moments. Angell et al. (2020) help distinguish between the two types of content: while the tagline "survival of the quickest" could be considered to be non-topical, normal content for the Mini JCW Roaster, the tagline "Beef: with a lot of horses hidden in it" that was actually used by the Mini JCW Roaster campaign represents content associated to real-time marketing. In this case, the content makes reference to the scandal featured at the time in international press involving meat adulteration after the detection of horse DNA in beef burger products by Ireland's Food Safety Authority. Companies recognize that by creating interesting, insightful, and entertaining content on a particular topic, they can encourage interaction with their audience (Kerns, 2014). In particular, real-time marketing is characterized as being cost-effective and affordable for large and small companies, a quick and concise way of communicating, and a way to humanize a brand and make it more entertaining (Kerns, 2014; Macy & Thompson, 2011).

Although real-time marketing is not a new concept, little academic research has been conducted on this topic in the marketing literature (Borah et al., 2020). This is particularly true regarding the specific consumer self-related mechanisms that underlie positive responses towards different real-time marketing contents. Despite contributing greatly to our understanding of real-time marketing strategies, those few studies that explore real-time marketing tend to focus on the real-time marketing content attributes that can lead to successful results. For example, Willemsen et al. (2018) analyzed whether different real-time marketing and nonreal-time marketing content differ in their potential to stimulate sharing behavior on social media and whether the originality and the meaningfulness of the message play a significant role. Nevertheless, the previous literature has shown that consumer responses to firm efforts are also shaped by elements intrinsic to the individual, such as involvement (e.g., Dessart, 2017). However, these elements have not yet been explored in the context of real-time marketing. Given that some of these elements are known to have a more profound and enduring effect on consumer responses (Dholakia, 2001), and since it is not yet known how their influence varies with distinct real-time marketing messages, more investigation is needed.

Consequently, based on the Elaboration Likelihood Model (Petty & Cacioppo, 1986) and congruence theory (Lafferty, 2007), the main objective of this study is to examine and understand how various audience-related variables (i.e., involvement, intensity of social media usage, self-brand congruence, and brand-moment congruence) and distinct real-time marketing contents (i.e., predictable and unpredictable contents) affect consumer responses, namely social media engagement. The relations are analyzed by resorting to the use of Structural Equation Modeling (SEM) and fuzzy-set qualitative comparative analysis (fsQCA). In this way, this study makes several contributions. For instance, at a managerial level, studying the effect of audience-related variables on engagement in distinct real-time marketing content contexts enables us to not only help managers understand the most valuable linkages overall, but also which are more expressive with predictable compared to unpredictable real-time marketing content. Based on the insights obtained, we propose a two-stage strategy regarding real-time marketing utilization to assist managers optimize their content usage in relation to audience-related characteristics. At a theoretical level, to the best of the researchers' knowledge, this is the first study that has been carried out to access the influence of individually related variables on engagement in different real-time marketing settings, and it thus extends the boundary of previous research and contributes significantly to engagement and real-time marketing literatures. Finally, at a methodological level, this study is the first to combine the above-mentioned methodological approaches in this field of research. Regression-based techniques based on net effects have typically been used in previous studies to analyse real-time marketing (e.g., Mazerant et al., 2021; Willemsen et al., 2018) and engagement (e.g., Carlson et al., 2018; Dessart, 2017), however the addition of fsQCA not only contributes to mitigate certain issues related with net-effects approaches, but it also provides a richer analysis of the phenomena under study (Wu et al., 2020). Accordingly, this study contributes at the managerial, theoretical, and methodological level.

2 | LITERATURE REVIEW AND HYPOTHESES

2.1 | Real-time marketing

The concept of real-time marketing has been discussed for more than 20 years, with various definitions being presented (Buhalis & Sinarta, 2019). Generally, and from a content standpoint, real-time marketing is defined as being the creation of branded content that is aligned with an event that is receiving greater public attention at the actual point in time, with the aim of capturing some of that visibility for the brand (Kerns, 2014). From a marketing automation perspective, some authors use the term real-time marketing to refer to the personalization of goods, services, or brand communications, in real time, based on consumer interests (Kallier, 2017; Macy & Thompson, 2011). In the journalism literature, real-time marketing is commonly associated with concepts such as newsjacking. Newsjacking involves the creation of branded content with implicit or explicit reference to current news in slogans and/or images (Angell et al., 2020). Despite the similarities between newsjacking and real-time marketing, real-time marketing also includes current events and online trends (Kerns, 2014), which are not always newsworthy.

Firms tend to attain higher performance when the corporate usage of social media is intense and the number of social media updates is high (Ribeiro-Navarrete et al., 2021). With the rise in popularity and usage of social media platforms, the notion of real-time marketing as a social media strategy has recently converged (Willemsen et al., 2018). Previous studies have identified specific functions which are transversal to all social media types, such as presence, sharing, relationships, or identity (Felix et al., 2017). The surge of augmented reality filters on social networks has created new user experiences that potentiate user satisfaction and sharing (Ibáñez-Sánchez et al., 2022), while virtual reality has generated new forms of presence for individuals and firms (i.e., immersive and vivid telepresence) (C. Chen & Yao, 2022). The construction of most of these functions is facilitated by the responsiveness provided by social media platforms for consumers (Mazerant et al., 2021). This responsiveness is reflected in the conversations per se, which demonstrate that consumers mostly engage in conversations regarding emerging, new topics that are taking place at that moment (Syrdal & Briggs, 2018). On the other hand, firms can also be responsive and can take part in the conversation by aligning their contents to the trends, topics, or events discussed at that moment on social media (Buhalis & Sinarta, 2019). By associating the brand content with timely, top-of-mind moments, brands can increase the topically and relevance of their messages for consumers and enhance the consumers' appreciation of their brand (Mazerant et al., 2021). According to De Vries et al. (2012), interactive and vivid brand posts result in an increase in the number of likes. With real-time marketing, and the underlying active social media listening (Saura, 2021), firms demonstrate marketing agility in the advertising area, where the firm rapidly understands emerging market trends and subsequently implements decisions to adapt to them (Kalaignanam et al., 2021).

Although firms increasingly adopt real-time marketing in an attempt to encourage more organic and positive consumer responses (Buhalis & Sinarta, 2019), "little is known about whether real-time marketing is indeed an effective social media strategy and, if so, why" (Mazerant et al., 2021, p. 15). Furthermore, the few empirical studies that examine the effect of real-time marketing messages on social media focus on those attributes directly related with the real-time marketing content that can potentially lead to successful results. For example, Willemsen et al. (2018) assessed the effect of various real-time and nonreal-time marketing contents on sharing behavior in social media, and evaluated how these relationships are mediated by the originality and meaningfulness of the message. On the other hand, by focusing on the engagement rate of consumers, Mazerant et al. (2021) extended the study of the influence of originality and meaningfulness to also include the creative crafting of the real-time marketing message. In general, creative advertising is associated with more favorable cognitive, affective, and behavioral outcomes (Darley & Lim, 2022). Finally, Borah et al. (2020) examined the effect of improvised marketing interventions (a similar interpretation to real-time marketing) on virality and firm value, and showed that humor and timeliness (or un-anticipation) accentuate the effect.

Notwithstanding the importance of content-related dimensions, in the previous literature, various individually intrinsic elements, such as self-brand congruence, involvement, or co-created value, have also been shown to mould consumers' responses to firm efforts (e.g., De Vries & Carlson, 2014; Dessart, 2017). In many cases, such as that of involvement (Dholakia, 2001), these elements may have a more profound and enduring effect on consumer responses than transient situational elements. Therefore, to narrow this research gap, this study examines and compares specific individual-related factors that underlie positive consumer responses in different content strategy contexts. This understanding is founded on both the Elaboration Likelihood Model and congruence theory.

2.2 | Theoretical background and hypotheses

Significant research has been conducted over the last decades in an attempt to explain changes in the capacity of distinct messages to persuade individuals to respond more positively to advertising at a cognitive, affective, and behavioral level (Scholten, 1996). Recently, attention has been given to content on social media (e.g., Colicev et al., 2018). In particular, the Elaboration Likelihood Model (Petty & Cacioppo, 1986) is a widely-applied theory to describe why certain forms of communication are more capable of persuasion (and are, consequently, more effective) than others. This model is a dual-process model that predicts changes in consumer attitudes towards advertised brands (Scholten, 1996), whereby these changes depend on both the audience and the context in which the advertised message is delivered (Angell et al., 2020).

The Elaboration Likelihood Model postulates that the response to a message is a result of the motivation and ability of individuals to elaborate on the message (Schmidt & Spreng, 1996). When an individual has the motivation and ability to process new information systematically, that individual is more likely to engage more deeply in message-relevant thinking (i.e., elaboration), and thus follows a central processing route (Lee & Kim, 2016; Petty & Cacioppo, 1986). According to Angell et al. (2020), individuals with more awareness, understanding, and predisposition towards current affairs are better placed and more motivated to navigate this process. Involvement with the brand, offering, or other features depicted in the message, are strongly related with the central route (Petty & Cacioppo, 1981), as more involved individuals are expected to retrieve and process more information (Colicev et al., 2018). Typically, messages processed through the central route produce stronger and more favorable evaluations and responses (Angell et al., 2020; Petty & Cacioppo, 1981).

On the contrary, when an individual lacks the motivation and the ability to thoroughly process the message, they show a lower message elaboration level, and thus tend to adopt a peripheral route (Lee & Kim, 2016; Petty & Cacioppo, 1986). The peripheral route relies more on the association of the elements of the message with the brand, the heuristic inference of brand quality from the elements of the message, or the attitudinal change through mere brand exposure (Scholten, 1996). The previous literature argues that elaboration via the peripheral route tends to have limited effects on further behaviors, since only short-term changes in attitude are produced, and thus content designers prefer their communications to be processed and elaborated by individuals in a deeper and more thoughtful way (Angell et al., 2020).

However, following the biased-elaboration postulate, the Elaboration Likelihood Model assumes that elaboration and consumer responses are influenced by the sense of congruency/incongruency informed by initial attitudes toward the brand (Scholten, 1996). Congruence theory suggests that the storage and retrieval of information from memory is affected by similarity, and that the more congruent, the better the association and consequent retrieval (Lafferty, 2007). Besides assessing the congruence of the brand with the individual self (Kressmann et al., 2006), individuals also tend to evaluate the congruence with the advertising claim (Drengner et al., 2011). At this stage, changes in attitude across the central route can have an upward or downward bias depending on the presence or absence of fit between the message and the initial attitude of the individual, which consequently affects the consumer's response (Scholten, 1996). Based on the general premises of the Elaboration Likelihood Model, we propose the conceptual framework of Figure 1 to study the effect of audience-related dimensions and content strategies on consumer engagement.

2.2.1 | Engagement as a central consumer response

In social media marketing, engagement has been highlighted as being one of the most important consumer responses (e.g., Ashley & Tuten, 2015; Syrdal & Briggs, 2018). Consumer engagement is a concept whose foundations are based on relationship marketing (Dessart et al., 2015) which has been covered extensively by several authors (Brodie et al., 2013). Despite its importance to researchers, different definitions, focuses, and dimensions of consumer engagement exist in the current literature, particularly in the context of social media marketing (Syrdal & Briggs, 2018; Vinerean & Opreana, 2021).

According to Vivek et al. (2012), consumer engagement denotes the intensity of the participation in and connection with the offerings or activities of an organization by individuals, whether it is the firm or the individual who initiates the contact. For Brodie et al. (2013), consumer engagement is “a context-dependent, psychological state characterized by fluctuating intensity levels that occur within dynamic, iterative engagement processes” (p. 107). Focusing on online brand communities, Dessart (2017) considers engagement to be “the state that reflects consumers' positive individual dispositions towards the community and the focal brand” (p. 377). Although subjacent in the former studies, Vinerean and Opreana (2021) explicitly show that engagement is a multidimensional construct that is manifested through cognitive, affective, and behavioral dimensions beyond transaction situations. Schivinski et al. (2016) further expand the behavioral dimension by arguing that consumers engage specifically at three levels: consumption (i.e., passive consumption without participation), contribution (i.e., intermediate level that includes interaction with brand content through likes, comments, shares, and other actions), and creation (i.e., highest level that implies more personal engagement through posting brand-related content).

2.2.2 | Predictable and unpredictable real-time marketing contents

Content strategies are based on the type of content chosen (Mazerant et al., 2021). Contrary to nonreal-time marketing contents (i.e., normal or regular contents [Mazerant et al., 2021]), real-time marketing contents are, by definition, intrinsically dependent on

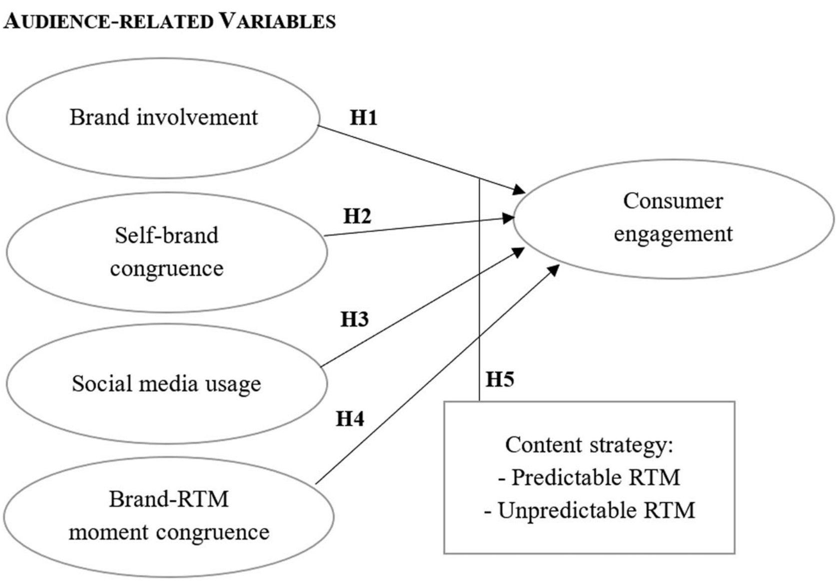

FIGURE 1 Conceptual framework

timely and public moments (Buhalis & Sinarta, 2019). In turn, these moments can vary greatly, giving rise to distinct typologies. For example, Kerns (2014) classifies real-time marketing moments as: planned real-time marketing (i.e., moments of large, public events where the topic and date of the event are both know in advance, such as the Oscars); watchlist real-time marketing (i.e., moments where the topic is known, but the date of the event is not, such as the birth of a royal baby); opportunistic real-time marketing (i.e., microtrends or topics that unexpectedly occur within a larger known event, such as an incorrect nomination in the Oscars); and everyday real-time marketing (i.e., both event and topic are unexpected, such as those that appear on Twitter Trending Topics).

Willemsen et al. (2018) propose another typology. These authors argue that social media content can be classified based on the predictability of the moment. Predictable moments are those that can be foreseen since they regularly reoccur (e.g., Christmas) or are announced in public calendars (e.g., the Super Bowl). These moments gather anticipated public attention, and thus brands can create and plan real-time marketing content strategies in the medium or long term. On the other hand, unpredictable moments are those that are not announced. They simply occur and are impossible to anticipate, thus requiring brands to adapt quickly (e.g., Oreo took advantage of an unexpected power outage during the Super Bowl of 2013 and tweeted "Power out? No problem" with a visual of an Oreo cookie in the dark and the tagline "You can still dunk in the dark"). For simplicity purposes, this study uses the classification of Willemsen et al. (2018) and subdivides real-time marketing content into predictable and unpredictable real-time marketing.

The previous literature shows that the type of social media content employed can affect consumer responses. For example, Tafesse (2015) concluded that posts with a higher degree of novelty and consistency positively contributed to the number of likes and shares on social media. Considering the real-time marketing context specifically, Willemsen et al. (2018) found that social media messages that use real-time marketing content, because of their focus on current happenings, evoke more shares when compared to those messages that do not use real-time marketing. Furthermore, the authors showed that real-time marketing messages are an even more effective content strategy when the messages are associated with unpredictable, rather than predictable events. The authors argue that this occurs owing to consumers' increased focus on and processing of newsworthy content, which creates the need for brands to adapt to a more journalistic role.

2.2.3 | The effect of audience characteristics

The Elaboration Likelihood Model also assumes that consumer responses are dependent on audience characteristics (Angell et al., 2020). Based on the premises of this theory, this study proposes and tests involvement, intensity of social media usage, self-brand congruence, and brand-moment congruence as potentially relevant factors in explaining engagement.

Involvement represents the personal interest and relevance of a focal object or decision in terms of values, goals, and self-concept (Hollebeek, 2011). Previous studies have found that more involved consumers gather more extensive information, exhibit more interest to process the information, and demonstrate more exploratory behaviors (De Vries & Carlson, 2014). Regarding the relationship of marketing constructs, involvement is considered to be a required antecedent before the expression of a customer's relevant engagement level (Hollebeek et al., 2014; Vinerean & Opreana, 2021).

Intensity of social media usage refers to the amount of time individuals actively spend on social media on a typical day (Ellison et al., 2007). Kumar et al. (2016) refer to users who show a greater predisposition and willingness to interact on these platforms as being "social network-prone" consumers. The authors argue that those consumers who are more inclined to use social networks value more the opportunity to connect with other customers of the brand in question. In turn, this constitutes a strong motivation for frequent social media presence and persuasion, and it can thus increase the receptivity of individuals to engagement on social media platforms.

Self-brand congruence refers to the correspondence between the consumer's self-concept and the brand's image or personality as perceived by the consumer (De Vries & Carlson, 2014; Kressmann et al., 2006). According to De Vries and Carlson (2014), when participating in contemplative evaluations and cognitive processing, consumers identify some brands as being relevant to their lives and congruent with their self-concept. If the correspondence between self-concept and brand image is high, then consumers express more brand preference and loyalty (Kressmann et al., 2006). Similarly, high levels of self-brand congruence can also lead to high social media engagement (De Vries & Carlson, 2014).

Finally, brand-moment congruence can be defined analogously to event-brand congruence. Event-brand congruence, which can also be called "event fit," refers to the degree of perceived similarity between the attributes of the brand and the attributes of the event (Drengner et al., 2011). According to Drengner et al. (2011), fit occurs when the consumer's knowledge of the brand matches their knowledge of the event. The similarity in attributes among the brand and the event/moment influences the storage and retrieval of information in memory, whereby the greater the congruency, the better the linkage and retrieval (Lafferty, 2007). Although very limited research exists regarding fit in the real-time marketing context, in the brand alliance field it has been argued that alliances between two brands with similar images elicit more positive responses from consumers (Pinello et al., 2022). On the contrary, partner image dissimilarity in relation to industry scope, firm size, and country-of-origin negatively affects the perception of brand fit (Decker & Baade, 2016). However, moderate dissimilarity can result in more favorable alliance evaluations as brands tend to complement each other's lacking attributes (van der Lans et al., 2014).

Based on the previous discussion, this study proposes that these audience-related variables are relevant for explaining engagement overall. Accordingly, for all the content strategies, we advance the following set of hypotheses:

H1: *Involvement with the brand has a positive effect on engagement*.

H2: *Self-brand congruence has a positive effect on engagement.*

H3: *Intensity of social media usage has a positive effect on engagement*.

H4: *Congruence between the brand and the real-time marketing moment has a positive effect on engagement.*

2.2.4 | The effect of audience characteristics by content strategy

Although audience-related variables are expected to affect engagement positively, it is possible that this can occur differently, based on the type of content encountered. According to a key principle of the Elaboration Likelihood Model, persuasion is tendentially higher when the recipients have a higher ability and greater motivation to address the arguments of the message (Scholten, 1996). Nevertheless, real-time marketing content, in particular, thrives on topicality, entertainment, and personal relevance (especially unpredictable content), which in turn greatly increases the attention and motivation of social media audiences, resulting in a higher sharing behavior (Willemsen et al., 2018). Therefore, those who are more interested, more involved, and who possess a greater knowledge of current affairs can be better positioned and more predisposed for a more deeper and a more thorough information processing, which results in more favorable attitudes and responses (Angell et al., 2020).

Additionally, individuals create and protect beliefs regarding their own identities, preferences, habits, and lifestyles, and patronize brands and brand content that reflect them (Kressmann et al., 2006). Recently, Syrdal and Briggs (2018) found that humorous content and up-to-date, relevant, breaking news stories (mostly related with unpredictable real-time marketing content) are the most engaging for consumers, as they more accurately mirror their self-image in terms of interests in politics, sports, or local news. Similarly, brands which are known for posting timely news, events, and trends that are consistent with the brand image may benefit from more consumer engagement, given the greater level of brand-moment congruence. According to the Elaboration Likelihood Model, individuals tend to bolster and support messages that are congruent with their initial attitude towards and image of the brand (Scholten, 1996).

Bearing these finding in mind, audience-related characteristics are expected to be particularly salient for engagement in the real-time marketing setting where unpredictable content is presented, when compared to the setting where predictable content is presented. When compared with predictable content, unpredictable content demonstrates the rapid creative ability of the firm regarding the use of current and catchy moments that are interesting and intrinsically relate with the consumer (Willemsen et al., 2018), which can consequently potentiate the link between, for example, consumer involvement (Angell et al., 2020) or self-brand congruence (Syrdal & Briggs, 2018) and engagement. The following hypothesis is thus proposed:

H5: *Audience-related characteristics are more relevant in explaining engagement with unpredictable real-time marketing content than with predictable real-time marketing content.*

3 | METHODS

3.1 | Data collection

The target sample consists of individuals who are over 18 years old and live in Portugal, who have an active social media account and are familiar with the brands under study (prior brand knowledge is a prerequisite for studying the consumer-brand relationship). The analysis was limited to Facebook and Instagram, as these comprise the top three most-used social networks in Portugal (Hootsuite, 2019). A convenience sample of 421 respondents was obtained from three online surveys (one for each brand). All the questionnaires comprised three sections with the same questions, differing only with regards the brand and the set of visual content displayed. In the first section, consumers were questioned about their social media usage and their general relationship with the brand (i.e., involvement and self-brand congruence). The remaining sections questioned the level of consumer engagement, based on two different visual contents. The visual content consisted of predictable and unpredictable real-time marketing posts published by each brand on social media during 2020 (Appendix A). These sections also questioned moment recognition and congruence between the brand and the real-time marketing moment.

The questionnaires were simultaneously distributed through various social media groups and also through the researchers' personal contacts during December 2020. A pretest of the questionnaire was performed through personal interviews with 15 respondents to assess the understandability of the questions. The sample comprises 58.2% females, 59.6% aged up to 24-year-old, 55.3% being undergraduates, 50.6% were employed or self-employed, and 62.9% earned a gross monthly income of 1000€ or less. Given the skewness of the sample mainly in terms of age and its potential to affect consumer behavior on social networks, the study analyses the measurement invariance (configural and metric invariance) of the constructs by age. The results support metric invariance (Appendix B - Table B1) and therefore the weights reported by respondents in the two age groups can be significantly compared.

3.2 | Brand selection

To decrease the effect of brand bias, the study jointly analyses three brands. The brands selected for the study were brands that regularly post real-time marketing content on their social networks, based on both predictable and unpredictable moments. For the selection of the brands, the rankings of Marketeer (https://marketeer.sapo.pt/—Marketeer Awards) and Briefing (https://www.briefing.pt/—Efficacy

Awards) from 2017 to 2020 were analyzed. Among the 255 nominations, three brands from the food and nonfood retail sectors which regularly upload real-time marketing posts were selected, namely: Lidl (a supermarket chain of German origin), Control (the gel, lubricant, and condom brand of the Artsana Group), and Licor Beirão (a spirits brand owned by the J. Carranca Redondo group). 19.0% of participants responded to the Lidl questionnaire, 41.8% to the Control questionnaire, and 39.2% to the Licor Beirão questionnaire.

3.3 | Measures

All the measures used in this study were adapted from the previous literature (Table 1). Involvement and self-brand congruence were adapted from De Vries and Carlson (2014), and brand-moment congruence from Drengner et al. (2011). Each item was measured with a seven-point Likert scale, ranging from (1) totally disagree to (7) totally agree. To measure engagement, the study adapted the items used by Schivinski et al. (2016), applying a seven-point Likert-type scale, ranging from totally improbable (1) to totally probable (7). The study measured engagement with the two different visual contents of each brand by using the same scale (Appendix A). Finally, the ordinal categories to measure social media usage were drawn from Ellison et al. (2007).

Confirmatory factor analysis (CFA) allows us to validate construct measurement properties. The CFA model presents good fit to the data ($\chi^2 = 328.502$, $df = 139$, $p = 0.000$, CFI = 0.968, TLI = 0.957 and RMSEA = 0.057), considering the usual cut-off values, as Kline (2011) suggests. The loading of each item relates only to one latent variable, which demonstrates the constructs' unidimensionality, as suggested by different authors (e.g., Ping, 2004). In line with Fornell and Larcker (1981), the constructs' internal reliability (Cronbach's $\alpha \geq 0.70$) (Table 1), composite reliability (CR ≥ 0.70), and average variance extracted (AVE ≥ 0.50) (Table 2) exceed the recommended levels, which accordingly confirmed the good reliability of the constructs. Moreover, for each latent variable, loadings are high, mostly above 0.80 (0.528–0.944), and significant at a level of 0.001, supporting the existence of convergent validity (Anderson & Gerbing, 1988; Hair et al., 2010; Ping, 2004). The analysis of discriminant validity confirms the low correlations between the constructs (Table 2), as supported in the CFA model. Furthermore, the square root of AVE values are higher than the correlation between the constructs, as stated by Fornell and Larcker (1981) and therefore all the measures of the constructs show good psychometric properties and the structural relations can be tested through the use of the research SEM model.

3.4 | Analysis models

The study considers various engagement responses as dependent variables, namely: (1) total engagement (Model 1: *engTOTAL*), irrespective of the content type, with $n = 421$; (2) engagement with predictable real-time marketing content (Model 2: *engRTM-P*), with $n = 169$; and (3) engagement with unpredictable real-time marketing content (Model 3: *engRTM-U*), with $n = 252$. The inclusion of participants in the groups/models depended on the results regarding real-time marketing moment recognition, which assesses the ability of individuals to decode the message, based on pre-existing knowledge from memory, as postulated in the Elaboration Likelihood Model (Petty & Cacioppo, 1986). Moment recognition was assessed based on the technique of Angell et al. (2020), which consists of: (1) directly questioning respondents whether they can identify the moment and, if so, (2) asking them to briefly describe it. Responses were then categorized as correctly or incorrectly identifying the moment, based on a list of acceptable terms (Appendix C - Table C1). Only 169 respondents correctly identified the predictable real-time marketing moment and 252 correctly identified the unpredictable real-time marketing moment. Since accurately evaluating brand-moment congruence depends on correct moment recognition, those respondents who were unable to correctly identify the real-time marketing moment were not considered for Models 2 and 3. The rigorous assessment of moment recognition also greatly contributes to guaranteeing the study's validity.

3.5 | SEM and fsQCA

This study uses both SEM and fsQCA to assess how specific audience-related variables relate to engagement in distinct real-time marketing settings. The purpose and guiding principles of each technique are different. We combine these methods for two main reasons. First, the consumer behavior literature, as a field-in-progress, continually supports the conceptualization and reconceptualisation of new relations and variables which are believed to exercise relevant influences on consumer behavior, in an attempt to advance theoretical understanding (Jacoby, 2002). We argue that by adding fsQCA, we can contribute with a richer analysis and, consequently enable more sophisticated models, relations, and analyses to emerge. Second, complementing SEM with fsQCA enables the study to mitigate some important limitations associated with regression-based models (Woodside, 2013), which are now discussed.

On the one hand, SEM is a multivariate technique used in scientific research to evaluate and test multivariate direct and indirect causal relations (Fan et al., 2016). Regression-based models, such as SEM, center around the principles of symmetry, linearity, unifinality, and additive and net effects of those independent variables that compete among themselves to explain variations in the dependent variable (Woodside, 2013). On the other hand, FsQCA is a set-theoretic technique which focuses on combinational effects between conditions (the independent variables) and outcomes (the dependent variables) (Fiss, 2011). FsQCA enables causal asymmetry (i.e., conditions that explain high outcome levels are not the exact opposites of those that explain low outcome levels) and also conjunctural causation (i.e., configurations of conditions can be necessary, sufficient, or both to obtain the outcome, although the

TABLE 1 Multi-item measures of dependent and independent variables.

Measures and Items	Cronbach's α Model 1	Model 2	Model 3
Involvement[a]: adapted from De Vries and Carlson (2014)	0.932	0.926	0.935
- This brand is important to me			
- I consider this brand a relevant part of my life			
- I am interested in this brand			
- I am involved with this brand			
Self-brand congruence[a]: adapted from De Vries and Carlson (2014)	0.929	0.933	0.925
- This brand is similar to me			
- This brand reflects the person I am			
- I identify with the image of this brand			
Engagement[b]: adapted from Schivinski et. al (2016)	0.840	0.842	0.837
Based on image _[c], indicate how likely it is that you would:			
- Follow the brand on social media			
- See the brand's other posts on social media			
- See the brand's website			
- Put a “like” on this post			
- Comment on this post			
- Share this post (publicly or privately)			
- Make a post about the content or brand on social media			
- Make a post about the content or brand on blogs or forums			
- Write brand recommendations online			
Brand-RTM moment congruence[a]: adapted from Drengner et al. (2011)	0.927	0.924	0.928
- There is a logical relationship between the topic and the brand			
- The topic and the brand images are similar			
- There is a correspondence between the topic and the brand			
- It makes sense for the brand to address the topic in its contents			
Measures and Items	**Frequencies Model 1**	**Model 2**	**Model 3**
Social media usage: adapted from Ellison et al. (2007)			
Average time spent on social media per day in the previous week:			
- Until 30 minutes	18	6	12
- Between 31 minutes and 1 hour	56	15	41
- Between 1h01 minutes and 1h30 minutes	77	37	40
- Between 1h31 minutes and 2 hours	92	38	54
- Between 2h01 minutes and 2h30 minutes	49	17	32
- Between 2h31 minutes and 3 hours	50	25	25
- More than 3 hours	79	31	48

[a]Likert scale of 1 = Totally disagree; 7 = Totally agree.

[b]Likert scale of 1 = Totally improbable; 7 = Totally probable.

[c]The image/content varied between planned RTM content (Model 2), and unplanned RTM content (Model 3). Model 1 is the total model. The images/contents for each brand are in Appendix A. The same scale was used in all contexts.

TABLE 2 Measurement evaluation of total engagement (engTotal)

		CR	AVE	*involv*	*self*	*moment*	*engTotal*
involv	Involvement	0.934	0.781	**0.88**			
self	Self-brand congruence	0.940	0.821	0.719	**0.91**		
moment	Brand-moment congruence	0.961	0.757	0.418	0.408	**0.87**	
engTotal	Total engagement	0.823	0.619	0.491	0.485	0.663	**0.79**
social	Intensity of social media usage	-	-	0.170	0.050*	0.245	0.272

Note: Square root of AVE values are in bold at the diagonal of the correlation matrices.

Abbreviations: AVE, average variance extracted; CR, composite reliability.

*Not significant at $p < 0.05$.

conditions are neither necessary, nor sufficient). This method also considers equifinality (i.e., distinct conditional configurations can result in a same outcome) and multifinality (i.e., the same conditions can result in distinct outcomes) (Pappas & Woodside, 2021; Ragin, 2008). This study assumes the audience-related variables to be the conditions, with the outcomes being the engagement levels in the three models. SEM is applied by using SPSS Amos and the fsQCA calculations are made using fsQCA 3.0 software (www.fsqca.com).

3.5.1 | FsQCA: Calibration

FsQCA is based on the idea of set membership, where raw data is transformed into fuzzy sets that range from zero (fully excluded from the set) to one (fully included in the set) (Ragin, 2008). Before starting calibration, an index score is created for all outcomes and conditions (except *social*) by calculating the average of the respective indicators. Next, following the calibration process used in Woodside (2013), three anchors are fixed to calibrate the data: (1) the 5th percentile of data values is the point of full nonmembership, (2) the 50th percentile of data values is the cross-over point, and (3) the 95th percentile of data values is the point of full membership. For the *social* variable, given the distinct measurement procedure, and to avoid misrepresenting cases (Pappas & Woodside, 2021), a direct theoretical anchor was established, where two represents full nonmembership, four represents the cross-over point, and six represents full membership. Table 3 presents the descriptive statistics and calibration values of each model. To prevent dropping cases from each analysis, 0.001 was summed to scores with the exact value of 0.5 (Fiss, 2011).

3.6 | Robustness check and predictive validity

The study carries out several calibration changes to assess the robustness of our models. First, we alter the thresholds for full membership and full nonmembership by subtracting 0.25 from each 95th percentile value and by adding 0.25 to each 5th percentile value, and repeat the analysis for the models (Afonso et al., 2018). The analysis produces the same results as those in Table 7, with the difference of models *engTOTAL* and *engRTM-U* having one more condition in one configuration. Second, in separate analyses, we alter the cross-over point in each model by adding and subtracting 0.25 to the 50th percentile value, and repeat the fsQCA tests (Santos & Gonçalves, 2019). Similar to the previous procedure, the results are consistent and have very few changes, with the existence of only small variations in the number of solutions and causal combinations. The few variations tend to appear when adding 0.25, however this led to no major alterations in interpretation. For example, the second configuration of *engTOTAL* does not appear. In general, the results support the robustness of our models.

This study also tests predictive validity. Predictive validity is crucial to show how well the model predicts the outcome with additional samples (Pappas & Woodside, 2021; Woodside, 2013). Since it is a general model, predictive validity is only examined for Model *engTOTAL* (Table 4). We start by splitting the sample into the modeling sample (1st subsample) and the holdout sample (2nd subsample). Next, highly consistent configurations are calculated for the 1st subsample, which are subsequently tested with the data from the 2nd subsample. This is carried out to assess the predictive capacity of the configurations for the holdout sample. We repeat the process for the 2nd subsample, using the data from the 1st subsample. The configurations of the 1st subsample demonstrate the existence of a high predictive capacity for the 2nd subsample, and vice-versa.

4 | RESULTS

4.1 | SEM results

The study estimates the full research SEM model (Table 5), M1: *engTOTAL*, to test the research hypotheses H1, H2, H3, and H4. The fit of the model is good, using the reference values of the fit indexes (e.g., Kline, 2011). All independent variables have a positive and significant effect on engagement ($0.120 \leq \beta \leq 0.510$; $p < 0.05$), except involvement, which is not significant. The research therefore does not verify H1, but supports H2, H3, and H4. More specifically, M1: *engTOTAL* explains 50% ($R^2 = 0.50$) of the engagement with social

TABLE 3 Summary data of dependent and independent variables

	Independent variables (outcomes)			Dependent variables (conditions)			
	engTOTAL	*engRTM-P*	*engRTM-U*	*involv*	*self*	*social*	*moment*
Model 1 ($n = 421$)							
Mean (or median)	3.140	-	-	3.963	3.641	4.000[a]	4.723
SD	1.253	-	-	1.619	1.504	-	1.594
Calibration values at							
95%	5.000	-	-	6.250	6.000	6.000[b]	7.000
50%	3.110	-	-	4.000	4.000	4.000[b]	5.000
5%	1.000	-	-	1.000	1.000	2.000[b]	1.750
Model 2 ($n = 252$)							
Mean (or median)	-	3.267	-	4.188	3.777	4.000[a]	4.864
SD	-	1.249	-	1.550	1.480	-	1.559
Calibration values at							
95%	-	5.176	-	6.250	6.000	6.000[b]	7.000
50%	-	3.220	-	4.250	4.000	4.000[b]	5.250
5%	-	1.000	-	1.250	1.000	2.000[b]	2.000
Model 3 ($n = 169$)							
Mean (or median)	-	-	3.054	3.813	3.549	4.000[a]	4.629
SD	-	-	1.251	1.650	1.515	-	1.612
Calibration values at							
95%	-	-	5.000	6.250	6.000	6.000[b]	7.000
50%	-	-	3.000	4.000	4.000	4.000[b]	4.750
5%	-	-	1.000	1.000	1.000	2.000[b]	1.750

Abbreviation: RTM, real-time marketing.

[a]The value concerns the median, since the variable is ordinal. The remaining values in this row are means.

[b]A direct method of calibration based on theoretical anchors is used in this case, instead of percentile values. The use of percentile values provides an inaccurate representation of the sample.

media content, with congruence between the brand and the real-time marketing moment ($\beta = 0.510$) being the most relevant explanatory variable. Self-brand congruence ($\beta = 0.180$) comes next, however its effect is much smaller and is fairly similar to that of intensity of use of social media ($\beta = 0.120$). Therefore it is fundamental to develop congruence between the brand and the real-time marketing moment to promote engagement with social media content.

To investigate the last hypothesis, H5, the research applies a multigroup analysis to evaluate measurement invariance (configural, metric, and structural invariance), considering two different real-time content marketing groups (M2: *engRTM-P*; M3: *engRTM-U*). Following Steenkamp and Baumgartner (1998) (Table 5), the study estimates the research model simultaneously in both groups without equality constraints (M0: Unconstrained model). The loadings are positive, with most of them being above 0.8, with the exception of two or three lower values, and they are significant at $p < 0.001$ (M2: 0.536–0.966; M3: 0.520–0.945). The existence of configural invariance is confirmed, as the model presents a good fit to the data ($\chi^2 = 499.385$, $df = 278$, $p = 0.000$, CFI = 0.963, TLI = 0.950 and RMSEA = 0.044) (e.g., Kline, 2011). The results indicate that the same construct has both the same meaning and the same structure in the different real-time marketing content groups. Next, the research proceeds to carry out the metric invariance analysis and estimates a second model (Ma: loadings equality, Table 6), imposing equality constraints between the loadings of the two groups (Steenkamp & Baumgartner, 1998). This model exhibits good fit to the data ($\chi^2 = 521.674$, $df = 127$, p-value=0.000, CFI = 0.962, TLI = 0.950, and RMSEA = 0.043) (e.g., Kline, 2011). The χ^2 difference test (Table 6) between the two models (M0–Ma) is nonsignificant ($\chi^2 = 22.289$, $df = 13$, $p = 0.051$), suggesting metric invariance. Furthermore, there is no difference in the model fit indicators ΔCFI = 0.000 and ΔRMSEA = 0.000 between the two models, meaning that all the loadings equality constraints hold, thus corroborating the evidence of metric invariance (F. F. Chen, 2007). Therefore, the weights reported by respondents in the two groups (M2 and M3) can be significantly compared. Carrying out the structural invariance evaluation

TABLE 4 Predictive validity

Causal combinations of the 1st subsample			Test of causal combinations of the 1st subsample using data from the 2nd subsample		
Causal Configuration	**Raw cov.**	**Cons.**	**Causal Configuration**	**Raw cov.**	**Cons.**
*self*social*	0.52	0.79	*self*social*	0.53	0.89
*involv*self*moment*	0.53	0.88	*involv*self*moment*	0.58	0.91
Solution coverage: 0.63			Solution coverage: 0.68		
Solution consistency: 0.80			Solution consistency: 0.87		
Causal combinations of the 2nd subsample			**Test of causal combinations of the 2nd subsample using data from the 1st subsample**		
Causal Configuration	**Raw cov.**	**Cons.**	**Causal Configuration**	**Raw cov.**	**Cons.**
*involv*self*	0.68	0.85	*involv*self*	0.63	0.78
*involv*social*	0.56	0.83	*involv*social*	0.57	0.79
*Social*moment*	0.55	0.84	*social*moment*	0.57	0.80
Solution coverage: 0.83			Solution coverage: 0.79		
Solution consistency: 0.77			Solution consistency: 0.75		

Note: "*" represents the logical operator AND.

TABLE 5 Engagement and real-time marketing models' fit and estimates (with SEM)

		Multigroup	
	M1: *engTOTAL*	M2: *engRTM-P*	M3: *engRTM-U*
Engagement <-- Social media usage	0.120*	0.099	0.070
Engagement <-- Involvement	0.130	0.139	0.118
Engagement <-- Self-brand congruence	0.180*	0.345**	0.044
Engagement <-- Brand-moment congruence	0.510**	0.329**	0.675**
R^2	0.52	0.44	0.62
Loadings (min–max)	0.528–0.944	0.536–0.966	0.520–0.945
RTM models Fit	M1: *engTOTAL*	M0: Unconstrained model	
χ^2	328.500	499.385	
df	139	278	
p Value	0.000	0.000	
CFI	0.968	0.963	
TLI	0.957	0.950	
RMSEA	0.057	0.044	

Note: All loadings are significant at $p < 0.001$.

Abbreviations: CFI, comparative fit index; *df*, degrees of freedom; RMSEA, root mean square error of approximation.

Regression weights:

*Significant at $p < 0.05$.

**Significant at $p < 0.001$.

(Steenkamp & Baumgartner, 1998) is the next step, which enables us to test research hypothesis H5. The study estimates a new model with additional regression weights equality constraints, namely Mb: regression weights equality. The comparison between the fit of the M0 and Mb models (Table 6) results in $\Delta\chi^2 = 54.231$, $\Delta df = 38$, and $p = 0.043$, which means that the structural coefficients are dissimilar between the two real-time marketing content groups and that the differences are significant (no structural invariance). Since ΔCFI = 0.001 and ΔRMSEA = 0.001, the differences between the two models (M0–Mb) are not very expressive. The research also addresses the CFI values of the individual models to choose which model is better and which are worse (Kline, 2016). Accordingly, the evidence points to structural coefficients variance.

To identify the source of this structural variance, the study estimates four new models, imposing equality constraints successively in each of the structural coefficients on the M2: *engRTM-P* and M3: *engRTM-U* models. Next, the study compares the χ^2 test and models fit measures between each of the four models and M0: Unconstrained model, that is, Mb1: b1 equality (*social*), Mb2: b2 equality (*involv*), Mb3: b3 equality (*self*), and Mb4: b4 equality (*moment*). By analysing Table 6, the nonsignificant $\Delta\chi 2$ test and the Δfit models' measures of Mb1: b1 equality (*social*) ($\Delta\chi^2$, $p = 0.798$) and of Mb2: b2 equality (*involv*) ($\Delta\chi^2$, $p = 0.910$) indicate the absence of statistically significant differences between the M2: *engRTM-P* and M3: *engRTM-U* models on the b1 (*social*: $\beta = 0.099$, $\beta = 0.070$) and b2 (*involv*: $\beta = 0.139$, $\beta = 0.118$) regression weights, respectively. However, the significant $\Delta\chi^2$ test and Δfit models' measures of Mb3: b3 equality (*self*) ($\Delta\chi^2$, $p = 0.034$) and of Mb4: b4 equality (*moment*) ($\Delta\chi^2$, $p = 0.002$) denote statistically significant differences in b3 (*self*) and b4 (*moment*) regression weights between M2: *engRTM-P* and M3: *engRTM-U* models. Indeed, Table 5 shows that the b3 (*self*) regression weights are $\beta = 0.345$ (M2: *engRTM-P*) and $\beta = 0.044$ (M3: *engRTM-U*;

TABLE 6 Real-time marketing multigroup models' comparison

Model	$\Delta\chi^2$	Δdf	*p* Value	ΔCFI	ΔRMSEA
Ma: Loadings equality	22.289	13	0.051	0.001	0.001
Mb: Regression weights equality	54.231	38	0.043	0.002	0.002
Mb1: b1 equality (*social*)	0.065	1	0.798	0.000	0.001
Mb2: b2 equality (*involv*)	0.013	1	0.910	0.000	0.001
Mb3: b3 equality (*self*)	4.515	1	0.034	0.000	0.000
Mb4: b4 equality (*moment*)	9.443	1	0.002	0.001	0.000

Abbreviations: CFI, comparative fit index; *df*, egrees of freedom; RMSEA, root mean square error of approximation.

nonsignificant), while b4 (*moment*) is $\beta = 0.329$ (M2: *engRTM-P*) and $\beta = 0.675$ (M3: *engRTM-U*). H5 is therefore partially corroborated, since only brand-moment congruence (*moment*) is more relevant in explaining engagement with unpredictable real-time marketing content compared to predictable real-time marketing content, thus registering an effect that almost duplicates. It should be noted that intensity of social media use and involvement do not add explanatory power in creating engagement, in both the cases of predictable and unpredictable real-time marketing content. Self-brand congruence only contributes to content engagement in the case of predictable real-time marketing content.

4.2 | FsQCA results

4.2.1 | Analysis of necessary conditions

According to Schneider and Wagemann (2010), the analysis of necessary conditions for the presence or absence of outcomes is the first step in fsQCA. In this study, no necessary conditions were found for each outcome (*engTOTAL*, *engRTM-P*, and *engRTM-U*) and corresponding absence (*~engTOTAL*, *~engRTM-P*, and *~engRTM-U*) at a consistency threshold of 0.90 (Ragin, 2000).

Analysis of sufficient conditions

The analysis of sufficient conditions starts with the constriction of the truth table (Pappas & Woodside, 2021). One truth table was calculated for each of the outcomes considered in this study when running the fsQCA software. The next step is to reduce the truth tables (Fiss, 2011). This reduction uses a recommended threshold of at least 80% of the cases in each sample (Ragin, 2008). In addition, the study uses a consistency threshold of 0.80 or above to identify which configurations are sufficient for each outcome (Ragin, 2008). The study then selects the intermediate solution of each outcome (Table 7), since this solution only considers simplifying assumptions consistent with empirical and theoretical knowledge (Pappas & Woodside, 2021). Given that consistency is greater than 0.74, and that coverage is between 0.25 and 0.65 for configurations and solutions, the models are considered to be informative (Woodside, 2013).

4.2.1.1 | *Causal configurations for the presence of outcomes*

The study starts by examining the causal conditions leading to the presence of the outcomes. The results shown in Table 7 indicate that three configurations explain total engagement (*engTOTAL*), three explain engagement with predictable real-time marketing content (*engRTM-P*), and two explain engagement with unpredictable real-time marketing content (*engRTM-U*). *EngTOTAL* occurs with high involvement (*involv*) paired with either high self-brand congruence (*self*) or high social media usage (*social*) and it can also occur with simultaneous high *social* and high brand-moment congruence (*moment*). On the other hand, *engRTM-P*, occurs with high *moment* combined with high *social*. It also occurs when *moment*, *self*, *involv* and are all simultaneously high, or *social*, *self*, and *involv* are all simultaneously high. Finally, *engRTM-U* occurs when *moment* is high or when *involv* and *self* are both high in combination.

4.2.1.2 | *Causal configurations for the absence of outcomes*

Possible causal conditions leading to the absence of outcomes are also examined. The findings in Table 7 show that three configurations explain the absence of engagement with the total sample (*~engTOTAL*) and with predictable real-time marketing (*~engRTM-P*), and that two configurations explain the absence of engagement with unpredictable real-time marketing (*~engRTM-U*). *~EngTOTAL* only occurs when *self* and *involv*, *social* and *moment*, and *self* and *moment* are all low in their respective combinations. On the other hand, *~engRTM-P* occurs with low *moment* combined with either low *self* or low *involv*. It also occurs with simultaneously low *self* and low *involv*. Finally, *~engRTM-U* occurs when *moment* is low and also when *involv*, *self*, and *social* are all low in combination.

As can be seen in the above-quoted models (presence and absence), most of the configurations show that the respective conditions that comprise these models need to be combined with others to be sufficient (except *moment* and *~moment* in *engRTM-U* and *~engRTM-U*). Accordingly, the conditions in each model are INUS conditions: "an *insufficient* but *necessary* part of a condition which is itself *unnecessary* but *sufficient* for the result" (Mackie, 1965, p. 245, emphasis in the original). Additionally, causal asymmetry can be established with the results, as most causal configurations that lead to the presence of the outcomes are distinct from the causal configurations that lead to their absence, and not merely the logical opposite.

5 | DISCUSSION AND CONCLUSION

In the social media sphere, firms are increasingly taking advantage of real-time marketing by associating their brand messages with public and real-time moments or events, with the objective of increasing the levels of brand image and consumer engagement (Mazerant

TABLE 7 Sufficiency analysis

Presence				Absence			
Model 1: *engTOTAL* = f (*involv*, *self*, *social*, *moment*)				**Model 1-absence:** *~engTOTAL* = f (*involv*, *self*, *social*, *moment*)			
Consistency cutoff: 0.83				Consistency cutoff: 0.86			
Causal Configuration	Raw cov.	Uni. cov.	Cons.	Causal Configuration	Raw cov.	Uni. cov.	Cons.
*self*involv*	0.65	0.16	0.82	*~self*~involv*	0.66	0.11	0.78
*social*involv*	0.57	0.02	0.81	*~social*~moment*	0.52	0.07	0.84
*social*moment*	0.56	0.08	0.82	*~self*~moment*	0.63	0.04	0.83
Solution coverage: 0.81				Solution coverage: 0.82			
Solution consistency: 0.75				Solution consistency: 0.75			
Model 2: *engRTM-P* = f (*involv*, *self*, *social*, *moment*)				**Model 2-absence:** *~engRTM-P* = f (*involv*, *self*, *social*, *moment*)			
Consistency cutoff: 0.86				Consistency cutoff: 0.82			
Causal Configuration	Raw cov.	Uni. cov.	Cons.	Causal Configuration	Raw cov.	Uni. cov.	Cons.
*moment*social*	0.55	0.12	0.81	*~moment*~self*	0.63	0.09	0.84
*moment*self*involv*	0.55	0.12	0.88	*~moment*~involv*	0.42	0.03	0.90
*social*self*involv*	0.53	0.10	0.86	*~self*~involv*	0.65	0.10	0.81
Solution coverage: 0.77				Solution coverage: 0.76			
Solution consistency: 0.81				Solution consistency: 0.77			
Model 3: *engRTM-U* = f (*involv*, *self*, *social*, *moment*)				**Model 3-absence:** *~engRTM-U* = f (*involv*, *self*, *social*, *moment*)			
Consistency cutoff: 0.80				Consistency cutoff: 0.89			
Causal Configuration	Raw cov.	Uni. cov.	Cons.	Causal Configuration	Raw cov.	Uni. cov.	Cons.
moment	0.78	0.24	0.77	*~moment*	0.76	0.35	0.77
*self*involv*	0.62	0.08	0.81	*~involv*~self*~social*	0.47	0.06	0.85
Solution coverage: 0.86				Solution coverage: 0.82			
Solution consistency: 0.75				Solution consistency: 0.76			

Note: "~" represents the absence of a condition and "*" the logical operator AND.
Abbreviation: RTM, real-time marketing.

et al., 2021). Although it is believed that real-time marketing is an effective strategy to promote word-of-mouth (Willemsen et al., 2018), little is known about how audience-related variables, such as involvement, self-brand congruence, or brand-moment congruence, contribute to engagement in the context of different real-time marketing contents. As such, the purpose of this study is to shed light on these relations, in an attempt to understand which audience and content characteristics are most likely to contribute to engage consumers.

Regarding the effects of each audience-related variable on engagement, the SEM results show that intensity of social media usage, self-brand congruence, and brand-moment congruence all contribute to explain consumer engagement, irrespective of the real-time marketing content considered (Model 1). On the other hand, the fsQCA results provide a more nuanced understanding of these variables, which could not have been evaluated with the exclusive use of SEM. As can be seen in the results, all audience-related variables (including involvement) are relevant to explain engagement, but they need to be combined differently among themselves to produce the outcome, which portrays the complexity mentioned by Jacoby (2002) when understanding consumer behaviors. Thus, if we observe involvement in isolation as a variable in competition with others to try to explain engagement (as with SEM), its role can be partially transferred to others that are possibly more expressive in the moment, as is the case of brand-moment congruence. However, with fsQCA, involvement is relevant, but only when considered simultaneously with other variables, such as self-brand congruence or intense social media usage. This result not only emphasizes the need for a configurational approach for engagement, but it also strengthens the idea that engagement can be achieved by multiple pathways (i.e., equifinality). In any case, the results of this study are generally in line with those of previous research. According to the Elaboration Likelihood Model, individuals who are more able and motivated (i.e., who are involved) are expected to participate in message-relevant thinking at a more profound level, retrieving and processing more information and thus producing stronger and more favorable evaluations and responses (Angell et al., 2020; Petty & Cacioppo, 1986). In addition, fit exists between self-concept and brand image, as well as between brand image and real-time marketing moment, when a similarity is perceived among each pair of variables (Kressmann et al., 2006). The higher this similarity, the higher the congruence, and thus the better the association and retrieval of information from memory, which consequently leads to the formation of more favorable attitudes (Lafferty, 2007).

When equating the possible effects of content strategy, the results show that the type of real-time marketing content employed does indeed matter. The SEM multigroup analysis detected statistically significant differences in the weight coefficients of brand-moment congruence between predictable (Model 2) and unpredictable (Model 3) real-time marketing content groups, although self-brand congruence was prominent in Model 2. In particular, self-brand congruence is more important in explaining engagement with predictable real-time marketing content, but brand-moment congruence has a far more expressive influence with unpredictable real-time marketing content. With fsQCA, the results are more nuanced, but are generally in line with this finding. In this case, with predictable real-time marketing, self-brand congruence and brand-moment congruence appear in two of the three configurations, in one of which their joint combination with involvement leads to higher engagement. Thus, although fit needs to exist between the brand and the predictable real-time marketing moment, it can be argued that, since the content is planned in advance regarding moments that are publicly anticipated, firms also need to be anchored in the connection that consumers have with the brand (and not only the predictable content) for engagement to occur, which is expressed by self-brand congruence. However, with unpredictable real-time marketing content, the fsQCA results indicate that brand-moment congruence alone can lead to higher engagement, although involvement and self-brand congruence combined are also important. Unpredictable real-time marketing contents are based on unexpected, catchy moments that are of interest for consumers (Willemsen et al., 2018). Willemsen et al. (2018) also found that these contents tend to be more meaningful than predictable content, and that they therefore result in more shares. If previous brand attitudes and current message fit closer, persuasion can be higher (Scholten, 1996), which accordingly influencing engagement considerably. For example, in the SEM results, it can be clearly seen that the effect of brand-moment congruence in the unpredictable real-time marketing context is double the effect of the same variable in the predictable real-time marketing context, where the remaining variables relegate their power to it.

Although the type of real-time marketing content is important, the relevance of audience-related variables should not be ignored, and this is more evidenced in the fsQCA results. In particular, although *moment* or *social* also have to be present in the configurations of Model 2, it is interesting to verify that *self*involv*, the highest consistent combination, is present across all models. The convergence of self-brand congruence with involvement is conceptualized by De Vries and Carlson (2014) as "brand strength," which was demonstrated to be key in consumer engagement formation. For these authors, as consumers advance in their assessment regarding the brand in question, some will assess the brand as being both significant to their lives and congruent with their self-concept, which highly strengthens their relationship with the brand, leading to higher engagement.

The fsQCA results also provide a unique complementary vision of the relations in question. For instance, all outcomes have a corresponding absence, however the configurations leading to them are combined differently. For example, *~self*~moment* (the absence) in Model 1-absence does not have a corresponding *self*moment* (the presence) in Model 1. Furthermore, in the case of low engagement with unpredictable real-time marketing content (*~engRTM-U*), low social media usage intensity (*~social*) appears, but *social* is not included in the corresponding high engagement model (*engRTM-U*) which denotes causal asymmetry that could not be assessed with SEM.

Finally, although the intensity of social media usage was not significant in explaining engagement in Models 2 and 3 with SEM, this condition was only found to be absent in Model 3 with fsQCA. However, it is worth noting that *social* normally appeared in the slightly lower consistency configurations, and that the standardized coefficient of this variable in SEM was among the lowest. Therefore, although it is relevant, the normalization of social media usage owing to its high penetration level can result in a marginally lower effect. Nonetheless, both the SEM and the fsQCA results indicate that with unpredictable real-time marketing content the intensity of social media usage is not important for engagement to occur. As such, we could be facing a situation where the "quality" associated with the content is more important than the "quantity" of time consuming it.

The findings of this study have important managerial implications, helping content managers understand the mechanisms that affect the formation of engagement in social networks, specifically in the context of different real-time marketing content, which is considered to be a necessary creative tactic to boost brand content on social networks (Macy & Thompson, 2011). However, first and foremost, even though most audience-related variables can help explain consumer engagement with social media individually, the results highlight that individuals inherently combine these variables differently, which has not been assessed in the previous literature. Accordingly, managers need to pay attention to the combinational effects of audience-related dimensions. In particular, this study reinforces the role of the consumer-brand relationship in engagement with content on social networks. Marketers need to design strategies that both reinforce the symbolic properties and values of the brand and also reflect the self-image of their consumers (N. J. De Vries & Carlson, 2014). Simultaneously, content managers should opt for topics that are easily recognized by the audience and that are highly congruent with the brand, which, otherwise, can cause irritation, confusion, and possible reputational loss (Angell et al., 2020; Macy & Thompson, 2011).

However, despite the attention that these variables deserve, managers can optimize the level of attention given to each of them, based on the type of real-time marketing content. As the study shows, unpredictable real-time marketing content leads to an engagement which is more content-dependent, while predictable real-time marketing content leads to engagement that is both content and audience-dependent. With this in mind, managers should consider a two-phased strategy. For example, involvement and self-brand congruence are both relevant dimensions for engagement formation, but they form at a deeper level (De Vries & Carlson, 2014), which can take time to solidify and produce results. Therefore, unpredictable real-time content which is congruent with the image of the brand should be used at a first stage to capture the attention of consumers more readily, which in turn can contribute to encouraging engagement more rapidly. This attention and positive disposition towards the brand is capable of posteriorly enabling the construction of stronger levels of consumer involvement and better self-brand congruence. When this is established, the strategy may enter a second phase, utilizing predictable real-time marketing content. As the results of this study indicate, even though brand-moment congruence continues to be important when using this content type, involvement and self-brand congruence are also necessary, as they represent elements that need to exist previously for engagement to occur.

This is not to say that content utilization should only be used in two consecutive moments, but rather that placing an emphasis on unpredictable real-time marketing content as a starting point can make a difference. In fact, we believe that unpredictable and predictable real-time marketing should be used interchangeably, since both also have disadvantages. For example, Borah et al. (2020) draw attention to the fact that preplanned contents can "lead a brand being seen as out of touch, distant from its target audience, and failing to capture the zeitgeist, or trends, feelings, and ideas that are typical at the time" (p. 88). On the other hand, unpredictable content cannot be planned in advance and requires more creativity, but previous literature shows that the time pressure associated with creative tasks can lead to ideas that are less original (Willemsen et al., 2018), thus creating pressure on the creative ability of marketing teams. One way to minimize this issue is to identify, hire, and delegate control of the message to individuals who are humorous and naturally tend to spot trends that can be included in the communication material and have the potential to go viral. However, since control is delegated and unplanned content requires quick responses, care should be given to avoid possible offensive content that could be posted too quickly, resulting in a backlash (Borah et al., 2020). Overall, to decrease the weaknesses of each content type, the use of both predictable and unpredictable content is recommended. However, regardless of the real-time marketing content used, it is suggested that visual formats are used, as these have been proven to increase sharing behavior (Willemsen et al., 2018).

In addition to assisting content managers, the findings of this study significantly contribute to expand the engagement and real-time marketing literature by developing and testing for the first time a theoretically driven model that relates different real-time marketing content strategies with audience-related variables to help explain social media engagement. Based on the Elaboration Likelihood Model (Petty & Cacioppo, 1986) and congruence theory (Lafferty, 2007), this study is the first to validate brand-moment congruence, self-brand congruence, involvement, and intensity of social media usage as relevant variables in explaining engagement with real-time marketing, and also to highlight the role of brand-moment congruence with the use of unpredictable real-time marketing. This validation is made by testing, also for the first time (to the authors' knowledge), the individual and combinatory effects of the proposed variables, and it thereby makes a useful methodological contribution. The fsQCA, in particular, permitted the evaluation of multifinality, equifinalty, and causal asymmetry, which could not have been assessed solely with SEM. Furthermore, while SEM showed a R^2 of 70%, 44%, and 62% for Model 1, 2, and 3, respectively, the solution coverages of these outcomes in fsQCA were 81%, 77%, and 86%, respectively, thus demonstrating a higher empirical relevance overall.

5.1 | Limitations and further research

This study has limitations that lead to the identification of suggestions for further research. The use of a convenience sample and dependent variables with distinct sizes presents representativeness issues. For example, the former resulted in the participation of very few respondents over 55 years of age, which does not allow conclusions to be drawn for more senior groups. Future research should thus use probability sampling procedures to obtain a sample which is more representative and more diverse. Since particular variables in this study vary according to the product categories and brands considered (e.g., involvement and self-brand congruence), replicating this study to other product category contexts and brands should also be pondered. Another limitation concerns the use of an engagement scale that centers on the behavioral element. Further studies should analyse engagement with real-time marketing as a three-dimensional construct including the cognitive and affective dimensions. Furthermore, attention should be given to control variables that were not included in this study, but which can have an effect on social media engagement, such as previous brand usage or involvement with the product, as well as contextual variables such as social media platforms and the influence of different online settings. Finally, although regression-based models have limitations, fsQCA also has its own boundaries. For example, according to Fainshmidt et al. (2020), the greater the number of conditions, the more complex the configurations and the more difficult it is to interpret them. Furthermore, fsQCA is dependent on the subjective decision-making of researchers during the design and implementation phases (e.g., calibration and threshold definition), which increases the risk for bias. As such, selecting the relevant variables is crucial, as well as determining a clear and exhaustive explanation of the design, reasoning, and analytical process.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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APPENDIX A

Brand posts selected for the study.

Brand	RTM content (based on a predictable moment)	Topic that inspired predictable RTM	RTM content (based on an unpredictable moment)	Topic that inspired unpredictable RTM
LIDL	 “It is not a realty, but it is a show!”	(Content published on September 26, 2020) Premiere of the reality show “Big Brother – A Revolução”, in September 2020[a].		(Content published on June 25, 2020) It is based on the mobile FaceApp that released a new update in June 2020 that allows people to visualize what their face would look like on the opposite sex[b].
CONTROL	 “After 1 PM, everything goes inside”	(Content published on November 14, 2020) Mandatory curfew starting at 1 PM on weekends determined on November 8 during the new state of emergency due to the COVID-19 pandemic[c].	 “When it is over and there is one that is still in denial”	(Content published on November 7, 2020) Donald Trump's denial on losing the USA election, in which Joe Biden was elected the new USA president on November 7, 2020[d].
LICOR BEIRÃO	 “Happy day of those who can only watch”	(Content published on June 1, 2020) Children's Day, celebrated annually on June 1.	 “Here for the curves. Congratulations, Miguel Oliveira!”	(Content published on November 22, 2020) Moto GP grand prix win by Portuguese motorcyclist Miguel Oliveira in Portimão on November 22, 2020[e].

Note. [a] TVI. (2020). "Big Brother - Revolução" já tem data de estreia. https://tvi.iol.pt/bigbrother/teresa-guilherme/01-09-2020/big-brother-a-revolucao-ja-tem-data-de-estreia

[b] Visão. (2020). FaceApp: A famosa aplicação russa voltou. https://visao.sapo.pt/atualidade/sociedade/2020-06-17-faceapp-a-famosa-aplicacao-russa-voltou-com-as-mesmas-letras-pequeninas-e-perigosas/

[c] Expresso. (2020). Governo impõe recolher obrigatório das 13h às 05h nos próximos dois fins de semana (mas o passeio higiénico é permitido). https://expresso.pt/coronavirus/2020-11-08-Governo-impoe-recolher-obrigatorio-das-13h-as-05h-nos-proximos-dois-fins-de-semana--mas-o-passeio-higienico-e-permitido-

[d] BBC News. (2020). Quem é Joe Biden, novo presidente eleito dos EUA. https://www.bbc.com/portuguese/internacional-54788165

[e] Visão. (2020). Miguel Oliveira é o vencedor do GP de Portugal de MotoGP. https://visao.sapo.pt/atualidade/desporto/2020-11-22-motogp-portugal-miguel-oliveira-venceu-gp-de-portugal-de-motogp/

APPENDIX B

The study analyses the measurement invariance (configural and metric invariance) of the constructs by age. Following the work of Steenkamp and Baumgartner (1998), the research considers two different age groups, namely the younger M1: = 25-year-old, and estimates a sequence of two nested models to evaluate if there are no statistically significant differences of the constructs between the two age groups. If this is the case, the weights reported by the respondents in the two groups can be significantly compared.

The results show evidence of configural variance, as the unconstrained model has good model fit to the data (e.g., Kline, 2016), as has already been explained in the main text of the paper, and the $\Delta\chi^2$ test between the unconstrained and loadings equality constrained models is nonsignificant ($p = 0.302$). Therefore, the results support metric invariance (Table B1).

TABLE B1 Measurement invariance by age groups

	Multigroup by age		
	M1: <25-year-old	M2: ≥25-year-old	
Loadings (min–max)	0.499–0.952	0.588–0.966	
R^2	0.44	0.61	
Models Fit	Unconstrained model		Δfit[a]
χ^2	540.14		15.081
df	278		13
p Value	0.000		0.302
CFI	0.957		0.000
RMSEA	0.047		0.000

Note: All loadings are significant at $p < 0.001$.

Abbreviations: CFI, comparative fit index; df, degrees of freedom; RMSEA, root mean square error of approximation.

[a]Differences between unconstrained vs. loadings equality constraints models.

APPENDIX C

See Table C1.

TABLE C1 List of accepted words regarding moment recognition, by content

Brand	RTM content	List of accepted and rejected words
LIDL	Predictable RTM content	Accepted words: "BB"; "Big Brother"; "BB 2020"; "BB Portugal"; "BB reality show"; "Reality show of TVI"; "BB—the Revolution"; "Big Brother the Revolution"; "Big Brother Reality show." Rejected words: Descriptions containing only the terms "Reality show" (for not specifying the program in question) and all references to reality shows other than Big Brother (e.g., House of Secrets) were rejected.
	Unpredictable RTM content	Accepted words: "Faceapp App"; "Face app application"; "application that changes the face of a woman to a man and vice versa/to change the sex/to see how we would look in the opposite gender/that shows the opposite sex/that changes to the opposite sex" (among others). Rejected words: "Instagram/Snapchat filters," "Face Swap," "mobile app"; "look mobile application" or "hair."
CONTROL	Predictable RTM content	Accepted words: "Recall/Gather/Isolation/Civic confinement/mandatory/after 1 PM/on weekends/after 1 PM on weekends"; "Restrictions/Measures/Rules imposed on circulation"; "Obligation to stay/stay at home after 13:00"; "Prohibition of circulation after 13:00" (among others similar). Rejected Words: Descriptions that mention only "Covid-19."
	Unpredictable RTM content	Accepted words: "Presidential/American/USA/United States elections"; "USA Election 2020"; "Loss/Fall/Defeat/Denial/Refusal of Trump in Presidential Elections"; "Loss/Exit of Trump"; "Joe Biden Win"; "Joe Biden New President"; "new USA president"; "Trump as ex-president" (among others related). Rejected Words: Descriptions that mention only "Trump."
LICOR BEIRÃO	Predictable RTM content	Accepted words: "Children's Day"; "World Children's Day"; "Children's Day on June 1st." Rejected words: "Father's Day"; "Under 18"; "Alcoholic Drink (+18 years)"; "Minors"; "Drink for over 18 years of age"; "Under 18."
	Unpredictable RTM content	Accepted words: "Portuguese/Portuguese Motorcyclist/Miguel Oliveira/MO/Portuguese Motorbike champion/winner/won the Moto GP" Vitória/Conquest/Congratulation/Celebration of Miguel Oliveira in the Moto GP"; "Moto GP"; "Miguel Oliveira" (among other similar words). Rejected words: "Formula 1"; "Motorcycles"; "Motard"; "Race Winner"; "Motorcycling."